Sumfree sets in groups

Terence Tao · 11 March 2016

https://terrytao.wordpress.com/2016/03/11/sumfree-sets-in-groups/

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Van Vu and I just posted to the arXiv our paper "sum-free sets in groups" (submitted to Discrete Analysis), as well as a companion survey article (submitted to J. Comb.). Given a subset A of an additive group $G = (G, +)$, define the quantity $\phi(A)$ to be the cardinality of the largest subset B of A which is *sum-free in* A in the sense that all the sums $b_1 + b_2$ with b_1, b_2 distinct elements of B lie outside of A. For instance, if A is itself a group, then $\phi(A) = 1$, since no two elements of A can sum to something outside of A. More generally, if A is the union of k groups, then $\phi(A)$ is at most k, thanks to the pigeonhole principle.

If G is the integers, then there are no non-trivial subgroups, and one can thus expect $\phi(A)$ to start growing with A. For instance, one has the following easy result:

> **Proposition 1** Let A be a set of 2^k natural numbers. Then $\phi(A) > k$.

Proof: We use an argument of Ruzsa, which is based in turn on an older argument of Choi. Let x_1 be the largest element of A, and then recursively, once $x_1, \dots, x_i$ has been selected, let x_{i+1} be the largest element of A not equal to any of the $x_1, \dots, x_i$, such that $x_{i+1} + x_j \notin A$ for all $j = 1, \dots, i$, terminating this construction when no such x_{i+1} can be located. This gives a sequence $x_1 > x_2 > \dots > x_m$ of elements in A which are sum-free in A, and with the property that for any $y \in A$, either y is equal to one of the x_i, or else $y + x_i \in A$ for some i with $x_i > y$. Iterating this, we see that any $y \in A$ is of the form $x_{i_1} - x_{i_2} - \dots - x_{i_j}$ for some $j \geq 1$ and $1 \leq i_1 < i_2 < \dots \leq i_j \leq m$. The number of such expressions $x_{i_1} - x_{i_2} - \dots - x_{i_j}$ is at most $2^m - 1$, thus $2^k \leq 2^m - 1$ which implies $m \geq k + 1$. Since $\phi(A) \geq m$, the claim follows. □

In particular, we have $\phi(A) \gg \log|A|$ for subsets A of the integers. It has been possible to improve upon this easy bound, but only with remarkable effort. The best lower bound currently is

$$\phi(A) \geq \log|A| (\log\log|A|)^{1/2-o(1)},$$

a result of Shao (building upon earlier work of Sudakov, Szemeredi, and Vu and of Dousse). In the opposite direction, a construction of Ruzsa gives examples of large sets A with $\phi(A) \leq \exp(O(\sqrt{\log|A|}))$.

Using the standard tool of Freiman homomorphisms, the above results for the integers extend to other torsion-free abelian groups G. In our paper we study the opposite case where G is finite (but still abelian). In this paper of Erdös (in which the quantity $\phi(A)$ was first introduced), the following question was posed: if A is sufficiently large depending on $\phi(A)$, does this imply the existence of two elements $x, y \in A$ with $x + y = 0$? As it turns out, we were able to find some simple counterexamples to this statement. For instance, if H is any finite additive group, then the set $A := \{1 \bmod 7, 2 \bmod 7, 4 \bmod 7\} \times H \subset \mathbf{Z}/7\mathbf{Z} \times H$ has $\phi(A) = 3$ but with no $x, y \in A$ summing to zero; this type of example in fact works with 7 replaced by any larger Mersenne prime, and we also have a counterexample in $\mathbf{Z}/2^n\mathbf{Z}$ for n arbitrarily large. However, in the positive direction, we can show that the answer to Erdös's question is positive if $|G|$ is assumed to have no small prime factors. That is to say,

> **Theorem 2** For every $k \geq 1$ there exists $C \geq 1$ such that if G is a finite abelian group whose order is not divisible by any prime less than or equal to C, and A is a subset of G with order at least C and $\phi(A) \leq k$, then there exist $x, y \in A$ with $x + y = 0$.

There are two main tools used to prove this result. One is an "arithmetic removal lemma" proven by Král, Serra, and Vena. Note that the condition $\phi(A) \leq k$ means that for any *distinct* $x_1, \dots, x_{k+1} \in A$, at least

one of the $x_i + x_j$, $1 \leq i < j \leq k+1$, must also lie in A. Roughly speaking, the arithmetic removal lemma allows one to "almost" remove the requirement that $x_1, \ldots, x_{k+1}$ be distinct, which basically now means that $x \in A \implies 2x \in A$ for almost all $x \in A$. This near-dilation symmetry, when combined with the hypothesis that $|G|$ has no small prime factors, gives a lot of "dispersion" in the Fourier coefficients of 1_A which can now be exploited to prove the theorem.

The second tool is the following structure theorem, which is the main result of our paper, and goes a fair ways towards classifying sets A for which $\phi(A)$ is small:

> **Theorem 3** Let A be a finite subset of an arbitrary additive group G, with $\phi(A) \leq k$. Then one can find finite subgroups $H_1, \ldots, H_m$ with $m \leq k$ such that $|A \cap H_i| \gg_k |H_i|$ and $|A \setminus (H_1 \cup \ldots \cup H_m)| \ll_k 1$. Furthermore, if $m = k$, then the exceptional set $A \setminus (H_1 \cup \ldots \cup H_m)$ is empty.

Roughly speaking, this theorem shows that the example of the union of k subgroups mentioned earlier is more or less the "only" example of sets A with $\phi(A) \leq k$, modulo the addition of some small exceptional sets and some refinement of the subgroups to dense subsets.

This theorem has the flavour of other inverse theorems in additive combinatorics, such as Freiman's theorem, and indeed one can use Freiman's theorem (and related tools, such as the Balog-Szemeredi theorem) to easily get a weaker version of this theorem. Indeed, if there are no sum-free subsets of A of order $k+1$, then a fraction $\gg_k 1$ of all pairs a, b in A must have their sum also in A (otherwise one could take $k+1$ random elements of A and they would be sum-free in A with positive probability). From this and the Balog-Szemeredi theorem and Freiman's theorem (in arbitrary abelian groups, as established by Green and Ruzsa), we see that A must be "commensurate" with a "coset progression" $H + P$ of bounded rank. One can then eliminate the torsion-free component P of this coset progression by a number of methods (e.g. by using variants of the argument in Proposition 1), with the upshot being that one can locate a finite group H_1 that has large intersection with A.

At this point it is tempting to simply remove H_1 from A and iterate. But one runs into a technical difficulty that removing a set such as H_1 from A can alter the quantity $\phi(A)$ in unpredictable ways, so one has to still keep H_1 around when analysing the residual set $A \setminus H_1$. A second difficulty is that the latter set $A \setminus H_1$ could be considerably smaller than A or H_1, but still large in absolute terms, so in particular any error term whose size is only bounded by $\varepsilon|A|$ for a small ε could be massive compared with the residual set $A \setminus H_1$, and so such error terms would be unacceptable. One can get around these difficulties if one first performs some preliminary "normalisation" of the group H_1, so that the residual set $A \setminus H_1$ does not intersect any coset of H_1 too strongly. The arguments become even more complicated when one starts removing more than one group $H_1, \ldots, H_i$ from A and analyses the residual set $A \setminus (H_1 \cup \ldots \cup H_i)$; indeed the "epsilon management" involved became so fearsomely intricate that we were forced to use a nonstandard analysis formulation of the problem in order to keep the complexity of the argument at a reasonable level (cf. my previous blog post on this topic). One drawback of doing so is that we have no effective bounds for the implied constants in our main theorem; it would be of interest to obtain a more direct proof of our main theorem that would lead to effective bounds.

Comments (11)

 **Anonymous** · 11 March 2016

Is it possible to get information on the structure of G from the knowledge of $\phi(A)$ for sufficiently many subsets A of G ?

 arch1 · 11 March 2016

"this paper of Erdos" – link missing?
[Corrected, thanks – T.]

 **anonymouse** · 12 March 2016

The link of Van Vu in the first sentence seems wrong.
[Corrected, thanks – T.]

 **HUMBERTO** · 13 March 2016

$order o(1) ???$

 **Rik Bos** · 4 December 2018

Does proposition 1 hold when A contains 0? For instance, if A={0,1,3,4} then k=2 and a sum-free subset of cardinality 3 does not exist.

 **Bik Ros** · 4 December 2018

0 is not a natural number

 **Rik Bos** · 4 December 2018

Well, that seems to be an ongoing debate. Since Bourbaki, many people tend to include 0 to the natural numbers. Also, this paper is about additive groups, where 0 plays an important role. Finally, a comment from https://terrytao.wordpress.com/books/analysis-i/ states:
"Natural numbers have no zero divisors" should read "Positive natural numbers have no zero divisors".

 **Bik Ros** · 4 December 2018

There should not be any debate. See the middle of page 7 of the paper he links to ("sum-free sets in groups", now named "sum-avoiding sets in groups"), which says "the natural numbers N = {1,2,...}".
With this in mind, the responses to your points are the following:
1). Terry does not consider 0 a natural number.
2). I don't see why 0 playing an important role should mean it is a natural number.
3). The example you provided seems more like an exception.

 **Rik Bos** · 6 December 2018

Well, especially the reference to p. 7 settles my question, even though the comment related to his book seems to indicate otherwise. Indeed, p. 18 of his book states:
Axiom 2.1. 0 is a natural number

 **Terence Tao** · 7 December 2018

Regarding the question of when to include zero amongst the natural numbers, I can refer readers to my previous comments on this question at https://terrytao.wordpress.com/books/analysis-i/#comment-439818 and https://terrytao.wordpress.com/2014/11/23/254a-notes-

1-elementary-multiplicative-number-theory/#comment-442582 (and also Remark 2.1.2 of my "Analysis I" book). The short answer is that in some contexts it is slightly preferable to consider zero a natural number, and in other contexts it is preferable not to. (But it is ultimately worth bearing in mind that, the quote attributed to Kronecker notwithstanding, the choice of what we do or do not consider to be a natural number is ultimately a human convention.)



Rik Bos · 7 December 2018

Thanks for your answer. So basically, you state that the inclusion or exclusion of 0 as a natural number is context-dependent. I guess my context is that 0 is a natural number unless stated otherwise (since as you write, it is ultimately a matter of convention).